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Key Points:

- Siliceous microzooplankton radiolarian records used to reconstruct the dynamic changes in the vertical structure of OT water since 20 ka
- Water changes at 0–500 m, 500–1,000 m, and 1,000–1,600 m in the OT under the impacts of the KC, North Pacific Intermediate Water (NPIW) and Pacific Central Water, respectively
- A rapid decline in NPIW influence at the Neoglaciation (~4–2 ka), similar to the KC, suggesting the high–low latitude linkage

Correspondence to:

L. Zhang and X. Zheng,
llzhang@scsio.ac.cn;
xufengzheng@hainanu.edu.cn

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Author Contributions:

Conceptualization: Lanlan Zhang
Data curation: Hu Chang, Lanlan Zhang
Formal analysis: Hu Chang, Lanlan Zhang, Yiping Yang
Funding acquisition: Lanlan Zhang
Methodology: Hu Chang, Lanlan Zhang
Resources: Xufeng Zheng
Supervision: Lanlan Zhang
Validation: Xun Gong, Rong Xiang
Writing – original draft: Hu Chang
Writing – review & editing: Lanlan Zhang, Xufeng Zheng, Xun Gong, Yiping Yang, Rong Xiang

Deglacial and Holocene Evolution of North Pacific Upper and Intermediate Ocean Circulation

Hu Chang[1,2,3], Lanlan Zhang[1], Xufeng Zheng[4], Xun Gong[5,6], Yiping Yang[1], and Rong Xiang[1]

[1]State Key Laboratory of Tropical Oceanography, Key Laboratory of Ocean and Marginal Sea Geology, South China Sea Institute of Oceanology, Chinese Academy of Sciences, Guangzhou, China, [2]School of Geographic and Oceanographic Sciences, Key Laboratory for Coast and Island Development of Ministry of Education, Nanjing University, Nanjing, China, [3]University of Chinese Academy of Sciences, Beijing, China, [4]State Key Laboratory of Marine Resource Utilization in South China Sea, Hainan University, Haikou, China, [5]Key Laboratory of Computing Power Network and Information Security, Ministry of Education, Shandong Computer Science Center, Qilu University of Technology (Shandong Academy of Sciences), Jinan, China, [6]State Key Laboratory of Physical Oceanography, Jinan, China

Abstract The interplay between the northward-flowing surface oceanic currents and the southward-flowing intermediate water masses regulated ocean stratification and mixing in the Northwest Pacific. Radiolaria, a biogenic proxy, was used to reconstruct changes in ocean stratification in the central Okinawa Trough (OT) paleoceanography since 20 ka. Our results revealed that changes in the shallow water (0–500 m), upper-intermediate water (500–1,000 m), and lower-intermediate water (1,000–1,600 m) are triggered by variations in the Kuroshio Current (KC), the North Pacific Intermediate Water (NPIW), and Pacific Central Water (PCW), respectively. The KC influence was stronger during the Holocene rather than during the last deglaciation, increased during the ~11.7–8.2 ka, weakened during the 8.2–4 ka, declined at 4–2 ka, and rapidly increased after 2 ka. In contrast, the influence of PCW increased during the last deglaciation, particularly during Heinrich Stadial 1 (HS1) and Younger Dryas (YD), but remained weak throughout the Holocene. This pattern appears to be closely linked to the Atlantic/Antarctic provenance events associated with Atlantic Meridional Overturning Circulation variability. Meanwhile, the influence of NPIW was strong during the YD but gradually weakened throughout the Holocene. Notably, a rapid decline in NPIW influence occurred during the Neoglaciation (~4–2 ka), similar to the KC, suggesting a high–low latitude linkage driven by atmosphere–ocean feedbacks mechanisms. This study provides robust data and a new perspective for understanding the connection to deglacial and Holocene variability in North Pacific upper and intermediate ocean circulation.

Plain Language Summary Although many studies of KC dynamics have been carried out, the change in KC influence during the Holocene is still debatable. Radiolarians are highly sensitive to changes in water mass properties (e.g., temperature and nutrient content). In this study, we analyze the high sedimentation rate and complete sedimentary history of the OT and reconstruct the vertical structure of the OT water and its evolutionary patterns by using characteristic radiolarian assemblage changes at shallow to deep-water depths, in order to clarify the dynamic changes in North Pacific upper and intermediate ocean circulations (KC, NPIW and PCW) since the last deglaciation on the basis of biogenic records from the OT.

1. Introduction

Oceanic circulations and their associated water masses play crucial roles in the transport of oxygen, nutrients, and heat (Saito, 2019). Changes in water mass structure and circulation patterns can significantly impact biogeochemical processes and the composition of marine organisms (Zhao et al., 2021). During the last deglaciation, the Earth experienced rapid climate changes, including the *Bølling and Allerød* warm phase (B/A, ~14.7–12.9 ka, a period of rapid climate warming characterized by increased global temperatures, modified oceanic current paths, and higher biological productivity in many regions) and the Heinrich Stadial 1 (HS1, ~17.5–14.7 ka, a temporary reversal to near-glacial conditions, with significant cooling in the North Atlantic and a weakened Atlantic Meridional Overturning Circulation (AMOC) and Younger Dryas (YD, ~12.9–11.7 ka, a prolonged cold event associated with massive iceberg discharges into the North Atlantic, leading to freshening and weakening of deep-water formation) cold periods (Kiefer & Kienast, 2005). These climatic transitions were associated with major reorganizations of global ocean circulation, including fluctuations in the AMOC, which had far-reaching effects on the Pacific Ocean via atmospheric and oceanic teleconnections (Zheng et al., 2016). To determine the water mass structure and its dynamic interactions with the climate system, it is essential to obtain robust data from the last deglacial period when the global climate and environment underwent significant changes.

The Okinawa Trough (OT), situated at the confluence of the Kuroshio Current (KC) and NPIW, provides a unique location to investigate how upper and intermediate ocean circulation responded to these climate shifts (Nakamura et al., 2013). Changes in KC influence heat and salinity transport into the North Pacific, while NPIW variability regulates intermediate water ventilation and nutrient cycling. Given that B/A, HS1, and YD were characterized by significant shifts in global climate and oceanic conditions, reconstructing the KC/NPIW interplay in the OT during these periods is essential for understanding the mechanisms linking tropical-subtropical Pacific circulation to high-latitude climate change. High-resolution paleoceanographic records from the OT can provide critical insights into the response of upper and intermediate water masses to abrupt climate events, helping to refine our understanding of Pacific circulation dynamics and their role in past climate variability.

Although many studies of KC dynamics have been carried out (Ding et al., 2022; Jian et al., 2000; Lee et al., 2024; Li et al., 2019, 2024; Pi et al., 2022; Wan et al., 2023; Xiang et al., 2007; Zhang et al., 2021), the change in KC influence during the Holocene is still debatable. Zhang et al. (2021) demonstrated a stronger KC during the Holocene, particularly based on alkenone-derived sea surface temperature (SST) records, which indicated a continuous warming trend. Building on this, Lee et al. (2024) further suggested that KC strength increased during the mid-to-late Holocene, as inferred from pollen-based reconstructions, which were linked to a weakened East Asian winter monsoon (EAWM). However, other proxy records, including grain size, planktonic foraminiferal assemblages, and the relative contribution of detrital ferrimagnetic minerals (RelDM), suggest a reduction in KC influence during the mid-Holocene (Ding et al., 2022; Li et al., 2019; Xiang et al., 2007). Additionally, diatom and Mg/Ca-based records indicate an intensified EAWM since the mid-Holocene (Li et al., 2024; Wan et al., 2023). Given that different proxies have varying sensitivities to seasonal changes and water depths, these discrepancies are more likely due to proxy biases rather than a fundamental contradiction in KC reconstructions. Moreover, these various proxies have their own specific implications and cannot recognize the information of vertical structures at shallow to deep-water depths. Therefore, understanding the evolution of ocean circulation requires multiple paleoceanographic proxies.

Single-cell microzooplankton, such as radiolarians, are highly sensitive to environmental changes and water mass properties. Their distribution and assemblage compositions are closely linked to water mass stratification, making them valuable indicators for reconstructing past oceanographic conditions in the OT (Qu, Xu, Wang, & Li, 2020; Qu, Wang, Xu, & Li, 2020; 2022). The geographical distribution of radiolarian species in the northwestern Pacific Ocean is significantly influenced by water masses, especially the KC, which plays an important role in the migration of shallow species (Zhang et al., 2018). Unlike plankton foraminifera, radiolarians with a siliceous skeleton can live in shallow to abyssal regions, and different species of radiolarians have unique ecological niches at their own depth. To better illustrate the necessity of using radiolarians to reconstruct water mass stratification changes in the OT, we first focus on shallow-dwelling warm-water radiolarians that serve as indicators of the KC. Species such as *Spongodiscidae*, *Dictyocoryne*, and *Tetrapyle* are associated with warm and oligotrophic surface waters transported by the KC (Qu, Xu, et al., 2020; Qu, Wang, et al., 2020). Their presence in high abundance suggests intensified KC influence in the OT. At the subsurface level, radiolarian species such as *Larcopyle buetschlii* is more abundant in transitional water masses, providing key information on variations in the thermocline depth and potential subsurface upwelling (Qu et al., 2022). These species are useful for detecting shifts in water mass interactions at intermediate depths. North Pacific Intermediate Water (NPIW)-related radiolarians, including *Larcopyle weddellium* and *Spongopyle osculosa*, serve as robust indicators of intermediate water changes (Matsuzaki et al., 2020). In addition, following Matsuzaki et al. (2022), we recognized the PCW-related radiolarian species, including *C. diadema, C. papillosum, C. profunda,* and *C. davisiana*. The nutrient-rich PCW is found at tropical depths of 1,000–2,000 m, which is a blend of Antarctic Bottom Water, Circumpolar Deep Water, and North Atlantic Deep Water origins, as noted from Talley (2007). Their variations in abundance reflect shifts in NPIW and PCW penetration and ventilation, which are crucial for understanding past water mass stratification in the OT. Previous research on radiolarian assemblages in the Sea of Okhotsk and the Bering Sea has demonstrated the effectiveness of such indicators in reconstructing intermediate water conditions (Itaki et al., 2008, 2012; Okazaki et al., 2003; Tanaka & Takahashi, 2005; Zhang et al., 2014), yet comparable studies in the subtropical North Pacific remain limited.

A recent study revealed radiolarian assemblage changes at shallow to deep-water depths in the subtropical Northwest Pacific (Matsuzaki et al., 2020). According to radiolarian assemblages used to reconstruct the evolution of water structure at various depths, previous studies have focused mainly on high-latitude regions (e.g., the Sea of Okhotsk and the Bering Sea) (Itaki et al., 2008, 2012; Okazaki et al., 2003; Tanaka & Takahashi, 2005; Zhang et al., 2014). In tropical and subtropical seas, however, knowledge of the evolution of water structure inferred from radiolarian assemblages is lacking.

In this study, we analyze the high sedimentation rate and complete sedimentary history of the OT and reconstruct the vertical structure of the OT water and its evolutionary patterns by using characteristic radiolarian assemblage changes at shallow to deep-water depths. By incorporating the ecological preferences of distinct radiolarian groups, we provide a comprehensive reconstruction of OT stratification changes since the last deglaciation. We aim to unravel the dynamic changes in North Pacific upper and intermediate ocean circulations (KC,NPIW and PCW) since the last deglaciation on the basis of biogenic records from the OT.

2. Materials and Methods

2.1. Chronology

Core Oki02 (4.92 m, 125.20°E, 26.07°N) was located in the central OT and was obtained in 2012 from a water depth of 1,612 m during the autumn open offshore cruise conducted by R/V *Kexue 1*, which is affiliated with the Institute of Oceanology, Chinese Academy of Sciences (Figure 1). Approximately 20 mg of mixed planktonic foraminifera was selected for accelerator mass spectrometry (AMS) ^{14}C dating, which was performed at the Beta Analytic Laboratory in the United States. The chronology of Core Oki02 was established using 14 AMS ^{14}C dates originally reported by Zheng et al. (2014) and Qian et al. (2022), which have been uniformly recalibrated to ensure consistency (Table 1). According to Qian et al. (2022), the surface sample (0–2 cm) AMS ^{14}C yielded a value of 100.6 ± 0.3 pMc, indicating a possible influence of nuclear explosions after 1950, so this age was not adopted (Table 1; Zheng et al., 2014). The other 13 AMS ^{14}C dates were converted to calendar years before present with the online program Calib 8.2 (http://calib.org) with the Marine 20 calibration data set, making adjustments for a regional ^{14}C reservoir age of 400 years and a ΔR of 70 ± 42 years (Yoneda et al., 2007). A continuous depth age model was constructed via the linear interpolation method to establish the age pattern of the entire core on the basis of the calibrated ^{14}C dates. The bottom layer of sediment has an age of approximately 19 ka. The high sedimentation rates and 1 cm sampling interval yield a temporal resolution of 20–50 years per sample.

We calibrated all the age models shown in Figures 3b and 4c following an identical treatment principle (the online program Calib 8.2 (http://calib.org) with the Marine 20 calibration data set). In addition, we calibrated the AMS ^{14}C ages of the other sequences cited in this study (without any raw data) and obtained small age differences that did not affect our conclusions about millennial-scale events.

2.2. Sample Processing and Species Identification

A total of 78 sediment samples were collected, with most samples taken from the core at 8-cm intervals and at 4-cm intervals during the Holocene. The sediments in the OT are exceptionally rich in radiolarians due to high primary productivity and favorable preservation conditions in this region. Previous studies have demonstrated that sediment samples from the OT yield abundant and well-preserved radiolarian assemblages, even when smaller sediment volumes are used (e.g., Matsuzaki et al., 2020; Zhang et al., 2018). The processing methodology proposed by Qiu, Zhang, Hu, et al. (2021), Qiu, Zhang, Xiang, et al. (2021), each sediment sample was weighed to approximately 0.1 g from a dry, raw sample to facilitate the identification of the relative quantities of radiolarian species. The weighed sediment was then subjected to sequential treatments with 10% H_2O_2 and 10% HCl solutions to remove organic matter and carbonates, respectively. The unscreened residues were mounted on glass slides measuring 24 mm × 50 mm (Zhang et al., 2009).

Radiolarian species were identified following taxonomical references (Chen et al., 2017; Chen & Tan, 1996; Matsuzaki et al., 2014, 2020; Tan & Chen, 1999; Zhang & Suzuki, 2017). At the species level, identifying more than 500 individuals from each sample was sufficient to obtain accurate species diversity values. At the abundance level, more than 1,000 individuals in each sample were counted to estimate radiolarian abundance (total individuals per gram dry weight, inds.g^{-1}). If the number of radiolarian specimens was below the mentioned limit, all the slides were checked. The absolute abundance was defined as the total number of radiolarians per

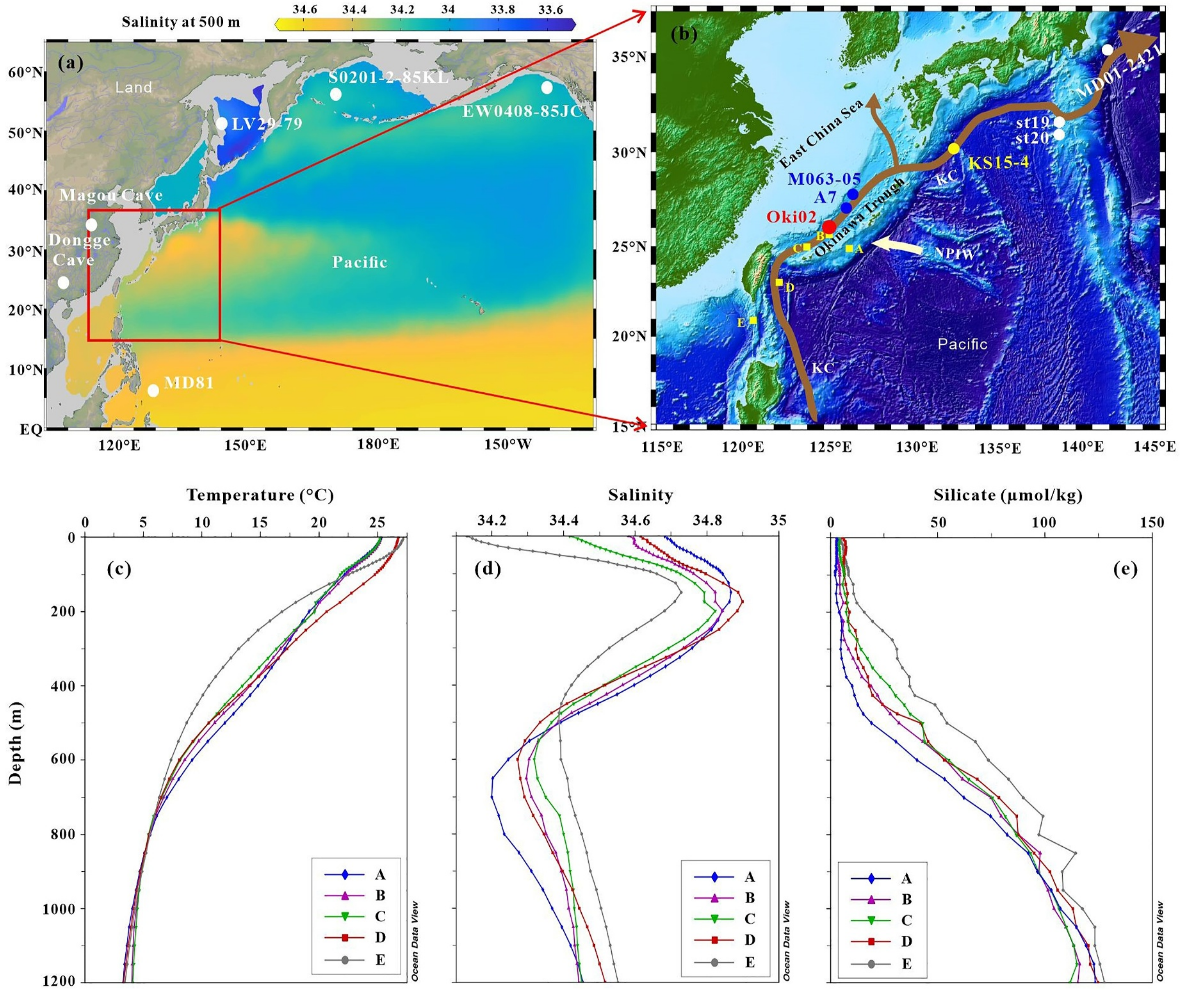

Figure 1. Regional setting (a) and the locations of Core Oki02 (this study), M063-05 (Li et al., 2019), A7 (Xiang et al., 2007), st19&st20 and MD01-2421 (Yamamoto, 2009), and Plankton-tow station KS15-4Sta.1 (Matsuzaki et al., 2020) (b). The vertical temperature (c), salinity (d), and silicate (e) profiles are from the World Ocean Atlas 2018 (Schlitzer & Sebastian, 2021). KC: Kuroshio Current; NPIW: North Pacific Intermediate Water.

gram of dried sample (inds·g^{-1}). Relative abundance is defined as the absolute abundance of a single species divided by the total abundance.

2.3. Radiolarian Assemblages at Different Depths

As marine microzooplankton living at deep depths, radiolarians exhibit distinct responses to variations in oceanic environmental parameters (Abelmann & Nimmergut, 2005; Itaki et al., 2012; Tanaka & Takahashi, 2005). Boltovskoy and Correa (2016) reported that temperature is the most important factor influeencing species distributions in the water column. In other words, temperature limits the species distribution at a certain depth in the water column. Furthermore, the abundance of these species assemblages at a certain depth can be regulated/controlled parameters, such as nutrients, in their habitat water conditions (Qiu, Zhang, Hu, et al., 2021; Qiu, Zhang, Xiang, et al., 2021; Zhang et al., 2009).

On the basis of the vertical distributions of living radiolarian species in the East China Sea and subtropical Northwest Pacific, as reported by Matsuzaki et al. (2016, 2020, 2022) and Okazaki et al. (2004), we identified radiolarian assemblages inhabiting depths of 0–500 m, 500–1,000 m, and 1,000–1,600 m. The shallow water (0–500 m) is characterized by *Peromelissa phalacra* Haeckel, *Theocorys turgidula* (Ehrenberg), *Botryocyrtis scutum* (Harting), *Phorticium pylonium* Haeckel, *Zygocircus piscicaudatus* Popofsky, *Acanthodesmia vinculatas* (Müller), *Pterocanium praetextum* (Ehrenberg), *Spongaster tetras* (Ehrenberg), *Didymocyrtis tetrathalamus*

Table 1
Accelerator Mass Spectrometry Radiocarbon and Calibrated Ages for Core Oki02 (Qian et al., 2022; Zheng et al., 2014)

Lab code	Depth (cm)	AMS^{14}C age (years BP)	Calibrated age (years BP)	2σ calibrated age range (years BP)
Beta-373510	0–2	100.6 ± 0.3pMc	–	–
Beta-337538	52–54	3,150 ± 30	2,709	2,498–2,874
Beta-373511	104–106	4,920 ± 30	4,935	4,754–5,196
Beta-373512	134–136	5,830 ± 30	5,974	5,782–6,171
Beta-337539	162–164	7,200 ± 40	7,433	7,265–7,580
Beta-373513	188–190	8,300 ± 30	8,541	8,367–8,753
Beta-373514	212–214	9,230 ± 40	9,709	9,502–9,954
Beta-337540	248–250	10,560 ± 40	11,546	11,286–11,803
Beta-571959	286–288	12,670 ± 40	14,038	13,791–14,293
Beta-571960	336–338	13,220 ± 40	14,976	14,697–15,235
Beta-337541	372–374	14,290 ± 50	16,367	16,093–16,650
Beta-571961	410–412	14,480 ± 50	16,616	16,336–16,901
Beta-571962	454–456	15,280 ± 50	17,596	17,310–17,889
Beta-337542	488–490	16,530 ± 60	18,967	18,722–19,240

Note. *pMc: percent modern carbon.

(Haeckel), *Dictyocoryne muelleri* (Haeckel), *Larcopyle* cf. *buetschlii* Dreyer, *Thecosphaera multispinula* (Su), *Druppatractus irregularis* Popofsky, and the *Tetrapyle* group, which form the shallow radiolarian assemblage. The upper-intermediate water (500–1,000 m) is characterized by *Larcopyle weddellium* Lazarus et al., *Spongopyle osculosa* Dreyer, *Flustrella* spp., which form the upper-intermediate radiolarian assemblage. The lower-intermediate water (1,000–1,600 m) is characterized by *Carpocanistrum diadema* Haeckel, *Peripyramis murrayana* (Haeckel), *Carpocanarium papillosum* (Ehrenberg), *Cornutella profunda* Ehrenberg, *Lithelius* spp., *Cycladophora davisiana* Ehrenberg, *Actinomma leptodermum* (Jørgensen), *Lithelius minor* Jørgensen, *and Lithelius nautiloides* Popofsky, which form the lower-intermediate radiolarian assemblage.

Furthermore, previous studies have shown that *D. tetrathalamus, Dictyocoryne elegans* (Ehrenberg), *D. muelleri*, and the *Tetrapyle* group are widely distributed in the warm pool region of the northwestern Pacific (Chang et al., 2003; Qiu, Zhang, Hu, et al., 2021; Qiu, Zhang, Xiang, et al., 2021, and references therein). Among them, *D. tetrathalamus* is particularly concentrated along the KC paths (Matsuzaki & Itaki, 2017). Therefore, we selected *D. tetrathalamus* as a key species to analyze the influence of the KC (Figure 2). KC influence in this study primarily as the relative strength of the KC in terms of volume transport, path and flow dynamics.

Previous studies have shown that the species *S. osculosa* and *L. weddellium*, which inhabit depths of 500–1,000 m, are found within the water mass of the NPIW (Matsuzaki et al., 2020; Okazaki et al., 2004). Therefore, we selected the assemblage of these two species as the NPIW assemblage to indicate the influence of NPIW.

At depths of 1,000–1,600 m, previous research has demonstrated that the tropical-subtropical species *C. diadema, C. papillosum, C. profunda,* and *C. davisiana* exclusively inhabit PCW (Matsuzaki et al., 2022). Consequently, we designated the assemblage of these four species as the PCW assemblage to reflect the influence of PCW.

3. Results and Discussion

3.1. KC Controls Change in Shallow Water in the OT With Variable Strength Over the Holocene

The shallow radiolarian assemblages can reflect the deglacial and Holocene variability in living-shallow-water conditions in the OT. Our results revealed that the absolute abundance of radiolarian species ranged from 446 to 14,700 inds.g^{-1} since 19.2 ka, with a total of 260 species. The relative abundances of the shallow radiolarian assemblage range from ∼10.5% to 37.0%, increased during the ∼11.7–8.2 ka, weakened during the 8.2–4 ka, declined at 4–2 ka, and rapidly increased after 2 ka (Figure 2). Similarly, as Figure 2 shown, the *Tetrapyle* group, *D. tetrathalamus*, and *D. muelleri* increased during the ∼11.7–8.2 ka, weakened during the 8.2–4 ka, declined at

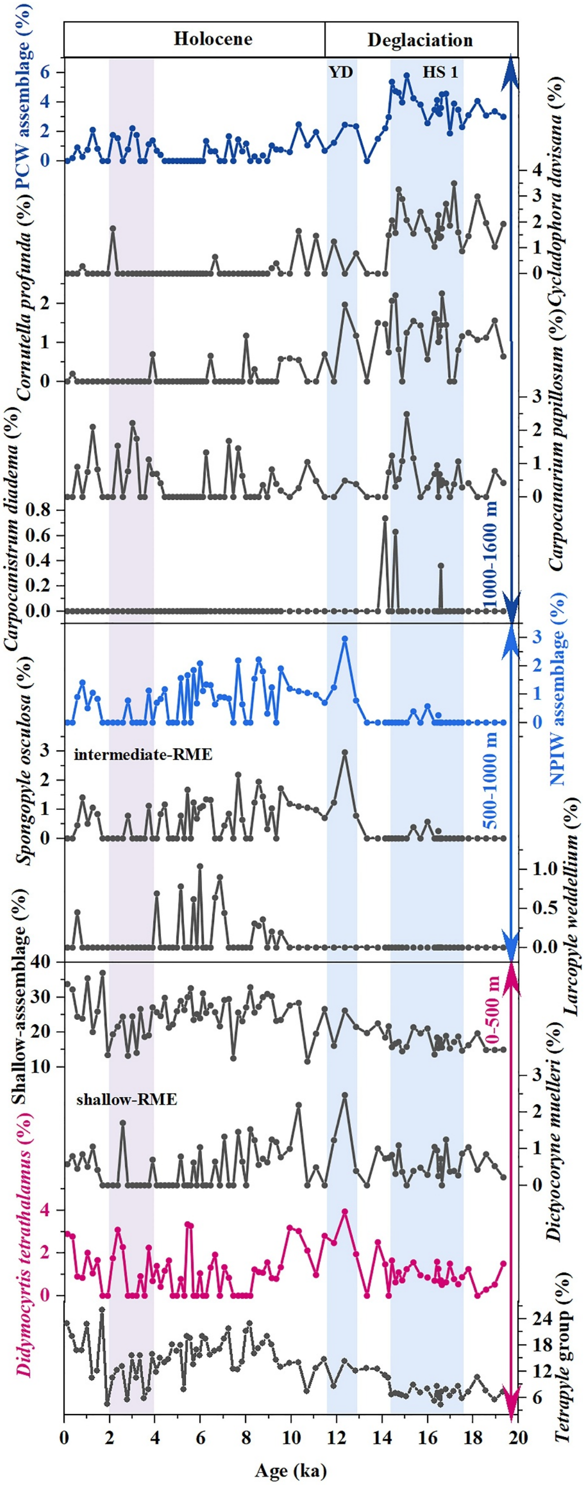

Figure 2. Relative abundances of radiolarians at different depths. The North Pacific Intermediate Water assemblage includes *S. osculosa* and *L. weddellium*, while the Pacific Central Water assemblage includes *C. diadema*, *C. papillosum*, *C. profunda*, and *C. davisiana*. YD: Younger Dryas; HS1: Heinrich Stadial 1; shallow-RME: shallow Radiolarian Minimum Event.; intermediate-RME: intermediate Radiolarian Minimum Event.

4–2 ka, and rapidly increased after 2 ka. The *Tetrapyle* group were dominant with high values (>6%) since 20 ka (Figure 2). Therefore, our data represent the KC influence gradually increased after the last deglaciation and reached a relatively high level during the early Holocene (~11.7–8.2 ka). During the mid-Holocene (8.2–4 ka), there was a declining trend in the KC influence, with the lowest weakening level occurring during the early late Holocene (4–2 ka), followed by a quick strengthening after 2 ka (Figures 3a and 4a). As a key indicator of KC, interestingly, the relative abundance of *D. tetrathalamus* has a high value during the YD.

Although many studies of KC dynamics have revealed a stronger KC during the Holocene than during the last deglaciation (Lee et al., 2024; Li et al., 2019; Pi et al., 2022; Wan et al., 2023; Xiang et al., 2007; Zhang et al., 2021), the changes in KC influence during the middle and late Holocene are still debatable due to the complicated physical and biogeochemical conditions caused by the combined influence of different water masses, which may lead to disagreements in paleoenvironmental interpretations (Dou et al., 2015; Kao et al., 2005; Li et al., 2018; Lim et al., 2017). During millennial periods, solar activity induces phase changes in the mean conditions of the tropical Pacific and governs centennial KC anomalies (Ding et al., 2022). Some proxies better capture what is happening at the surface close to 0 m, which is highly influenced by solar heat and thus insolation. By using organic geochemical data to reconstruct the temperature, Zhang et al. (2021) demonstrated a strengthened KC over the Holocene (including the mid-Holocene), peaking at approximately 4–3 ka, due to a continuous warming trend. Following Zhang et al. (2021), Lee et al. (2024) also reported that the KC was strengthened during the mid-Holocene and early late-Holocene by a weakened EAWM inferred from pollen-based temperature reconstruction. Recently, however, Li et al. (2024) and Wan et al. (2023) documented the weakened East Asian Summer Monsoon (EASM) and enhanced EAWM since the mid-Holocene on the basis of biogenic records (diatom and foraminifera), respectively, unlike the paleoenvironmental interpretation of Lee et al. (2024).

Moreover, multiple proxy records indicate a reduction in KC influence during the mid-Holocene and early late-Holocene. High-resolution grain-size records from the Okinawa Trough suggest that the KC weakened between 4.6 and 2.0 ka, followed by a rapid strengthening after 2.0 ka (Ding et al., 2022). This pattern is consistent with our radiolarian-based reconstruction. Notably, various proxies—including radiolarian (Figure 3a), grain-size data (Ding et al., 2022), the Relative Dissolution Method (RelDM), and the relative abundance of *Pulleniatina obliquiloculata*, a widely used indicator of KC strength—consistently suggest a weakened KC during the early late-Holocene (Li et al., 2019; Xiang et al., 2007; Figure 3b). Regarding the KC, some research primarily based on planktic foraminifera suggests that the current does not flow at the surface near 0 m but a bit deeper, around 50–150 m, and there is subsurface water related to the KC (Xiang et al., 2007). The *Pterocorys* group is distributed at water depths of 50–150 m (Zhang et al., 2018), showing a similar trend to *Pulleniatina obliquiloculata* between 4 and 2 ka, which helps clarify differences between indicators.

To distinguish between KC and East Asian Monsoon influences on sediment grain size, it is important to consider regional hydrodynamics and proxy-specific implications. While monsoon variability can influence sediment transport, previous studies have shown that fine-grained sediments in the Okinawa Trough are primarily controlled by KC dynamics rather than EAM-driven terrestrial input (Ding et al., 2022). Moreover, *Pulleniatina* *obliquiloculata* abundance fluctuations may be linked to variations in KC influence rather than monsoonal influences, as this species is a warm-water indicator transported by the KC (Li et al., 2019; Xiang et al., 2007). Although *Pulleniatina* events have been associated with winter monsoon variability in some studies, their occurrence in the OT aligns more closely with KC fluctuations. The combination of these proxies strengthens our interpretation that the observed changes primarily reflect KC variability rather than direct monsoonal effects.

The known Holocene *Pulleniatina* minimum event (PME) is of great concern, although its mechanism is unclear. Our shallow-water radiolarian record of *D. tetrathalamus*, a single-celled protozoon similar to foraminifera, further confirmed the reliability of the existence of the PME. Previous studies have suggested that PME events may reflect the weakening of KC activity, possibly as a result of the intensified winter monsoon (Giosan et al., 2018; Jian et al., 2000; Xiang et al., 2007). Furthermore, the decreased ratio of warm/cold water foraminifera indicates that the temperature becomes low, possibly associated with the weakened KC and the enhanced EAWM during the early late-Holocene (Figure 3c; Xiang et al., 2007; Wan et al., 2023). The significant decrease in the SST of the tropical western Pacific (Figure 3g; Stott et al., 2004) and the small temperature contrast in the midlatitude North Pacific (Figure 3e; Yamamoto, 2009) during the early late-Holocene (4–2 ka) may have contributed to the weakening of the KC.

The strengthening of the KC during the early Holocene may have been associated with the northward displacement of the Intertropical Convergence Zone (ITCZ) (Figure 3h). The ITCZ controls atmospheric convection and tropical cyclone (TC) activity, which can influence oceanic circulation. A northward shift in the ITCZ enhances moisture transport and convective activity in the western Pacific, potentially increasing TC formation (Sachs et al., 2018; Zhang et al., 2020). More frequent TCs can lead to stronger oceanic mixing and heat redistribution, which may contribute to an intensified KC (Cao et al., 2013; Contreras-Rosales et al., 2019). Additionally, ITCZ migration modulates the EASM, which, in turn, affects the KC through changes in wind-driven circulation and surface heat flux (Gray, 1979).

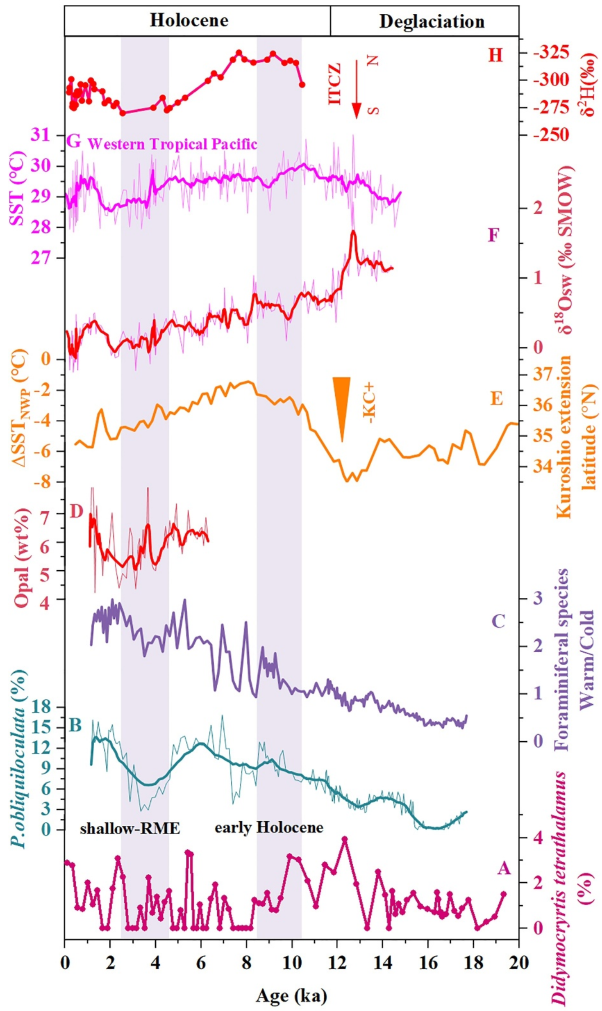

Figure 3. Variations in the relative abundances of the radiolarian KC indicator (*D. tetrathalamus*) in Core Oki02 (this study; a). The relative abundance of *Pulleniatina obliquiloculata* and variations in the warm/cold water foraminiferal species ratio in core A7 (Xiang et al., 2007; b, c). Opal (biogenic silica) contents of sediment core EW0408-85JC associated with Alaskan Current (Zhang et al., 2021; d). Latitudinal variation in the Kuroshio Extension as revealed by the sea surface temperature (SST) contrast between Core St.19 and MD01-2421 (Yamamoto, 2009; (e). Western tropical Pacific SST record and salinity index δ^{18}Osw (‰ SMOW) at site MD81 (Stott et al., 2004; f, g). δ^2H_{dino} from Palau T-Lake core PTLN-PC1 (Sachs et al., 2018; h). KC: Kuroshio Current; shallow-RME: shallow Radiolarian Minimum Event; ITCZ: Intertropical Convergence Zone.

3.2. NPIW and PCW Regulation of Intermediate Water in the OT

In this study, the relative abundance of the intermediate assemblage ranged from 0% to 20% and was generally higher than 5% during the last deglaciation and lower than 5% during the Holocene. NPIW and PCW radiolarian assemblages may present a change in the dwelling water depth (500–1,000 m) and (1,000–1,600 m) in the OT, respectively. In the upper-intermediate assemblage (500–1,000 m), *S. osculosa* and *L. weddellium* show coeval increases during the late deglaciation and early Holocene, reaching maximum abundances of ∼3.0% for *S. osculosa* and ∼0.1%–0.9% for *L. weddellium* (Figure 2). These peaks coincide with the reduction of PCW assemblage and are consistent with a reinvigoration of intermediate water ventilation following HS1. However, *L. weddellium* exhibits a broader temporal spread of elevated values from ∼9.5 to 5.6 ka, while *S. osculosa* shows a sharper and more transient peak around ∼12 ka, highlighting subtle differences in ecological preference or water mass affinity. In the lower-intermediate assemblage (1,000–1,600 m), the PCW-associated species show more differentiated patterns. *C. davisiana* dominates the group, with peak abundances of ∼3.5% during HS1 and YD, and a clear decline to <1% in the Holocene, signifying a strong link to glacial/interstadial deep water presence (Figure 2). *C. profunda* and *C. papillosum* also show increases during HS1 (∼1.0%–2.3% and ∼0.5%–2.5%, respectively), but unlike *C. davisiana*, their relative abundances remain slightly elevated during the early Holocene (∼1%–2%), suggesting a more gradual transition in deep-water properties. *C. diadema* displays the most muted variability, remaining

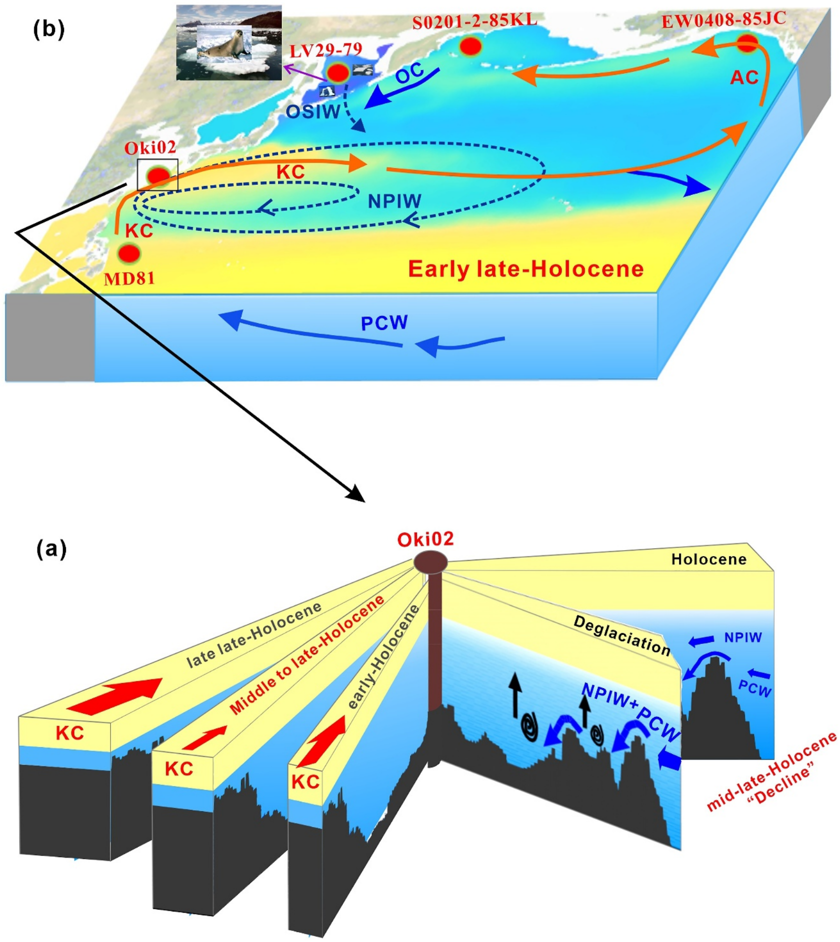

Figure 4. Illustration of deglacial and Holocene evolution of North Pacific upper and intermediate ocean circulation. (a) Changes in the KC, North Pacific Intermediate Water (NPIW) and Pacific Central Water (PCW) recorded by Core Oki02 since the last deglaciation; (b) schematic of the high–low latitude linkage for the decline in the KC, NPIW and PCW during the early late-Holocene (4–2 ka). AC: Alaska Current; OC: Oyashio Current.

below 1.0% throughout the record, which may reflect a broader tolerance or less distinct ecological niche compared to its counterparts. The NPIW assemblage reached its maximum value during the YD, followed by a gradual decline throughout the Holocene (Figure 5b). Similarly, the PCW assemblage peaked during HS1 and YD, with a notably weaker presence over the Holocene (Figure 5a).

The formation of sea ice in the mid- and high-latitude marginal seas of the North Pacific, such as the Sea of Okhotsk and the Gulf of Alaska, has led to the sinking of a dense shelf water mass accompanied by salt precipitation. This water mass mixes with the open ocean water mass along the iso-density surface, forming the oxygen-rich NPIW (Shimizu et al., 2004). Generally, the NPIW is distributed at depths between 300 and 800 m in the low-to mid-latitude Northwest Pacific (Yasuda, 1997). Although previous studies reported a significant increase in NPIW formation during the last deglaciation in the Okhotsk Sea and Bering Sea (Max et al., 2014), and highlighted the substantial influence of NPIW intrusion into the OT (Kubota et al., 2015; Zou et al., 2020), our data of NPIW species didn't indicate a particularly strong influence on the intermediate water of the OT (Figure 5b). Instead, an increase in NPIW occurred during the HS1 period (Figure 5b), which is consistent with the modeling results (Gong et al., 2019). Since the YD, the trend of NPIW species closely resembled the ventilation dynamics of the Okhotsk Sea Intermediate Water (OSIW) (Lembke-Jene et al., 2018) (Figures 5b and 5c). Additionally, the PCW influence during the last deglaciation was consistent with the $\delta^{13}C$ results and benthic foraminifera data from the Bering Sea, which are proxies for the NPIW variability (Max et al., 2014) (Figures 5a and 5c). According to modern observations, the majority of NPIW enters the OT through the Kerama Gap

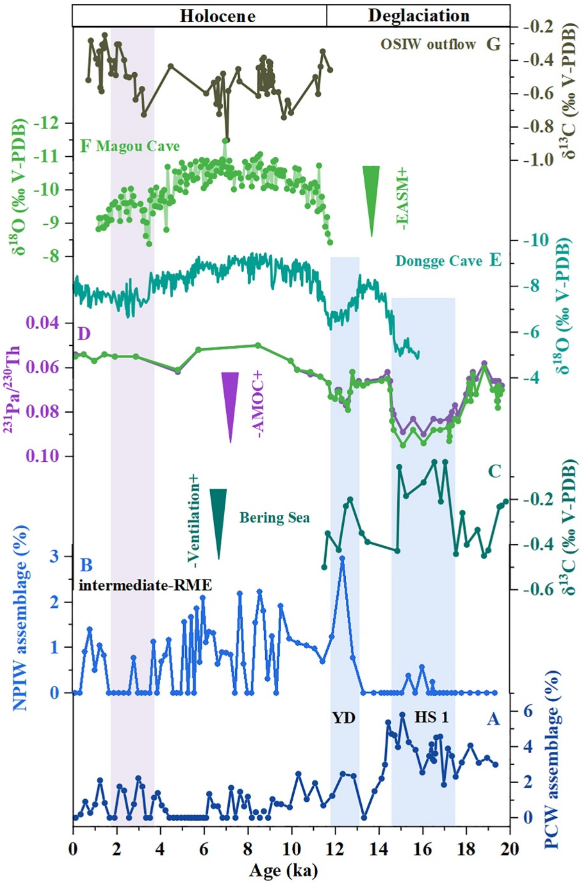

Figure 5. Variations in the relative abundance of the North Pacific Intermediate Water (NPIW) and Pacific Central Water radiolarian assemblages in Core Oki02 (this study; a, b); $\delta^{13}C$ as an NPIW index from the Bering Sea (Max et al., 2014; c); $^{231}Pa/^{230}Th$ as a proxy for Atlantic Meridional Overturning Circulation strength in the North Atlantic (McManus et al., 2004; d); Dongge Cave and Magou Cave speleothem $\delta^{18}O$ time series, which represents changes in the Southeast Asian summer monsoon (Dykoski et al., 2005, e; Cai et al., 2021, f); and the $\delta^{13}C$ record at site 79, which reflects Okhotsk Sea Intermediate Water ventilation dynamics (Lembke-Jene et al., 2018; (g). YD: Younger Dryas; HS1: Heinrich Stadial 1; intermediate-RME: intermediate Radiolarian Minimum Event.

(~1,100 m), and the intermediate water in the OT mixes to ~2,000 m depth (Nakamura et al., 2013). Additionally, during the last deglacial period, the bottom boundary of the NPIW was deeper than it is today, with its depth reaching over 2,000 m during HS1 (Jaccard & Galbraith, 2013; Matsumoto et al., 2002). As a result, there may have been a mixed influence of NPIW and PCW on the intermediate waters of the OT during this period. Furthermore, species that typically prefer the low salinity of NPIW may have been inhibited by the higher salinity of PCW, which could explain the lower values of NPIW-related species during this time (Figure 5b).

To further understand the mechanism behind the rapid increase in ventilation during HS1 and the YD, we compared the abundance of the NPIW and PCW assemblage with the $^{231}Pa/^{230}Th$ ratio, a proxy for the AMOC (McManus et al., 2004; Figure 5d). We found a clear inverse correlation between the intermediate assemblage and AMOC, indicating an Atlantic–Pacific seesaw. During HS1, catastrophic iceberg discharge occurred in the North Atlantic, causing weakening and even cessation of the AMOC (Gherardi et al., 2005; McManus et al., 2004). Paleoclimate studies have shown that the weakening or cessation of the AMOC resulted in a sharp decrease in temperature in most parts of the Northern Hemisphere on the scale of decades or even shorter periods (Thornalley et al., 2018). This cooling in the Northern Hemisphere promoted the formation of NPIW, and the NPIW and PCW assemblage provided evidence for the large-scale reorganization of NPIW and PCW during HS1 and YD, supporting the North Atlantic and North Pacific seesaw effects.

Unlike the north-south convective structure observed in the Atlantic Ocean, the North Pacific does not currently support deep-water formation. Instead, ventilation of the deep Pacific basin is primarily driven by the northward inflow of Antarctic Bottom Water. However, the existence of deep-water formation and Pacific Meridional Overturning Circulation (pMOC) during the deglaciation period remains a topic of debate (Okazaki et al., 2010). If such a circulation system had been active, an increase in NPIW/North Pacific Deep Water during HS1 and the YD would have likely enhanced upper-water circulation. The NPIW and CPW assemblage have a clear difference between HS1 and YD intervals. Namely, the PCW assemblage is higher during HS1 than YD, while the NPIW assemblage shows high only YD. This might be related to the difference in the ventilation depth that is previously proposed by Okazaki et al. (2010). During the YD, the KC indicator *D. tetrathalamus* also exhibits a distinct peak, coinciding with elevated abundances of *S. osculosa* (NPIW) and *C. profunda* (PCW), which can be interpreted as coincident environmental changes at different depths. This synchronicity could reflect a transient reorganization of Pacific water masses, possibly linked to a short-lived enhancement of the pMOC, as previously suggested by Okazaki et al. (2010).

Notably, the abrupt decrease in the intermediate assemblage (intermediate-RME) during the early late-Holocene (~4–2 ka) coincided with a decrease in NPIW in the western North Pacific (Lembke-Jene et al., 2018) (Figure 5g). This coincidence may indicate that the rapid decline in NPIW at 4–2 ka was responsible for the weakening of ventilation in the OT (Figures 5b and 5g).

3.3. A Synchronized Decline in the KC and NPIW in the Early Late Holocene

From the above results, we first recognized that there is a synchronized signal of the weaker KC and the weaker NPIW at ~4–2 ka on the basis of the unique microzooplankton of the OT records (Figure 4). At present, there are few records about the evolution of the Holocene NPIW. The $\delta^{13}C$ record of a deeper core (1,082 m) from the Okhotsk Sea, the foremost North Pacific ventilation region, revealed that a lower OSIW flowed during the early period of the late Holocene, suggesting significant weakening of the NPIW (Lembke-Jene et al., 2018). Moreover, the record of radiolarian *C. davisiana*, a proxy for the ventilation of the NPIW, indicates a significant reduction in OSIW at ~3.5–2 ka, coinciding with the cold Neoglaciation (Itaki et al., 2008). This decline suggests a weakening of intermediate-water formation in response to large-scale climatic changes.

During the early period of the late Holocene (~4–2 ka; Neoglaciation), millennial-scale variations in ocean circulation may have been linked to changes in atmospheric and hydrological conditions in the Northwest Pacific (Figure 4). Studies suggest that an intensified EASM led to increased precipitation in the Amur River catchment, contributing to enhanced freshwater discharge into the northwestern Okhotsk Sea (Cai et al., 2021; Dykoski et al., 2005; Herzschuh et al., 2019). This freshwater input may have influenced sea ice formation and water column stratification. Additionally, a weaker KC during this period (Li et al., 2024) may have reduced the northward transport of high-salinity water, potentially reinforcing stratification in the Okhotsk Sea (Figure 4). A decrease in upper water δ^{18}Osw (Stott et al., 2004) and biogenic silica (opal) accumulation (Davies et al., 2011) along the Alaskan coast further suggests possible reductions in ventilation and upwelling, which could have contributed to NPIW weakening.

Previous studies have shown that ENSO variability intensified during the mid-to-late Holocene, influenced by precessional solar forcing (Clement et al., 1999). ENSO-driven changes in atmospheric circulation and precipitation patterns could have affected hydrographic conditions in the Northwest Pacific, leading to reduced ventilation in the Okhotsk Sea and weaker NPIW formation. The synchronous variations observed in the upper- and intermediate-water circulation patterns in the OT may thus reflect the broader climatic influence of ENSO and its teleconnections with both low- and high-latitude climate systems.

4. Conclusions

This is the first study to use unique single-cell microzooplankton with siliceous shells to reconstruct the evolution of water structure in tropical-subtropical regions. Our data revealed an evolution of the multilayer structure in the OT water, including a shallow water layer from 0 to 500 m under the influence of the KC, an upper-intermediate water layer from 500 to 1,000 m controlled by the NPIW, and a lower-intermediate water layer from 1,000 to 1,600 m influenced by the PCW since 20 ka. The results of independent radiolarian records indicate that the KC in the Holocene was stronger than that during the Last Glacial Maximum and deglacial intervals, in contrast to the stronger-than-Holocene PCW under the last deglacial conditions. Furthermore, during the Holocene, the KC strengthened in the early period (~11.7–8.2 ka), weakened in the middle period (~8.2–4 ka), declined further between 4 and 2 ka, and then strengthened again in the late period (after 2 ka). This pattern may provide alternative evidence for the occurrence of the "*PME*" in the early late Holocene. In contrast, a strongly increased PCW occurred during the last deglaciation, especially during HS1 and YD, but remained weak over the Holocene, highly sensitive to the AMOC events. Meanwhile, the influence of NPIW was strong during the YD but gradually weakened throughout the Holocene. Interestingly, the NPIW influence also rapidly decreased during the early late Holocene (4–2 ka), possibly because of the sharply weakening KC. Thus, we propose a possible mechanistic explanation for the connection to high and low latitudes in the Northwest Pacific. During the Neoglaciation, under a background of weakened monsoonal activity and reduced brine rejection, the weakening KC may have contributed to enhanced stratification in the Sea of Okhotsk by reducing the northward supply of saline waters.

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Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

All new data presented in this article are openly accessible on Zenodo (Chang et al., 2025).

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